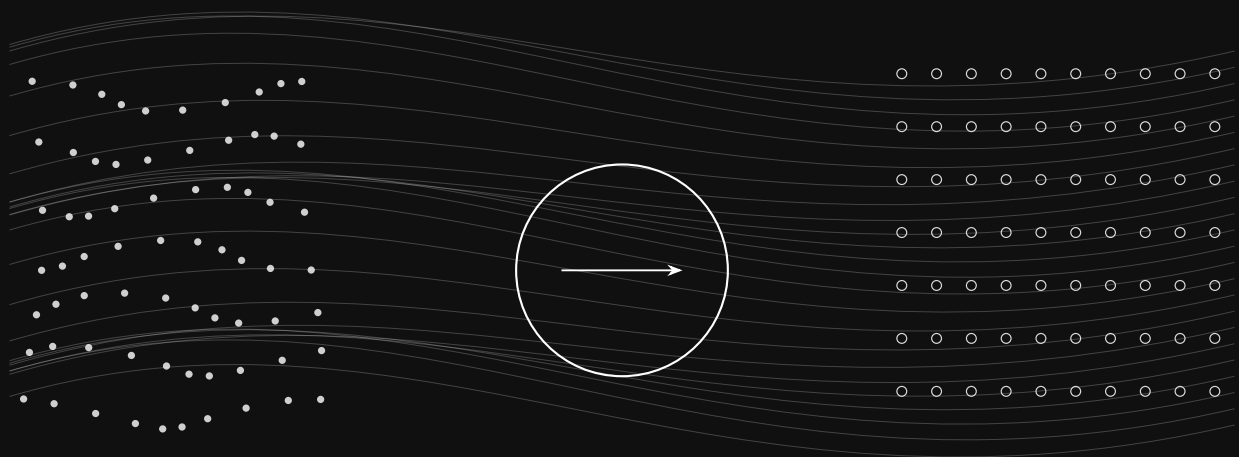


LITERATURE REVIEW

Requirements on Synthetic Data

for Training Machine
Learning Algorithms



OBSERVATIONS

REPRESENTATION

SYNTHETIC SAMPLES

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COURSE

Academic Research Skills
Literature review / Working draft

LITERATURE REVIEW / RESEARCH OVERVIEW

At a glance

ABSTRACT / TO BE WRITTEN

Summarise the problem, the scope of the literature search, the main findings and their implications for training machine learning algorithms. Complete this abstract after the review has been conducted.

KEYWORDS

Synthetic data / Machine learning / Data quality / Literature review

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ABOUT THIS WORKING PAPER

Prepared for Academic Research Skills, Faculty of Electrical Engineering, Mathematics and Computer Science, University of Twente. This is a designed manuscript scaffold: proposed questions, review dimensions and workflows are starting points, not research findings.

CONTEXT / RESEARCH DIRECTION

01 Introduction

This is the introduction to the research project.

1.1 Research background

Synthetic data can be used to train and evaluate ML algorithms.

Develop this background with cited evidence: explain the application context, the limitations of available real data and the motivation for using synthetic data.

PROPOSED RESEARCH QUESTION

What requirements should synthetic data meet to support the training of machine learning algorithms?

02 Literature review

This section reviews existing literature on synthetic data for machine learning.

2.1 Example citations

These references demonstrate the citation format and are starting points for reading; they do not constitute a completed literature synthesis.

Goodfellow et al. [1] introduce GANs. Conditional generation for tabular data is illustrated by CTGAN [2]. The review can examine fidelity and utility when comparing such approaches.

2.2 An organising framework

The dimensions below are proposed categories for structuring the review. Refine them using the selected studies; they are not validated findings.

Table 1. Proposed dimensions for extracting requirements from the literature.

Dimension	Guiding question	Evidence to extract
Fidelity	Which properties of real data must be preserved?	Distributional comparisons and domain constraints.
Utility	How is usefulness for the learning task assessed?	Downstream metrics and real-data baselines.
Coverage	Which groups and conditions need representation?	Subgroup results and rare-case analyses.
Privacy	How is disclosure risk evaluated?	Threat models, attack tests and stated guarantees.

2.3 Synthesis of the evidence

Compare where studies agree, where their requirements differ and how those differences depend on the task, data modality and evaluation setting. Support each substantive claim with a reference.

APPROACH / FROM SEARCH TO SYNTHESIS

03 Methodology

Describe the review procedure here. The workflow below is a proposed structure; update it to reflect the work actually performed.



Figure 1. Proposed literature-review workflow. No search or screening counts are implied.

3.1 Search strategy and selection

Record the databases, search date, exact search strings and inclusion and exclusion criteria. Explain how duplicate records and disagreements were handled.

3.2 Extraction and comparison

For each selected study, capture its task, dataset, generation method, stated requirements, evaluation metrics and limitations. Use Table 1 as an initial extraction structure.

04 Conclusion

Answer the research question using the completed synthesis. Distinguish requirements supported across studies from those that depend on a specific task or data setting.

SYNTHESIS / TO BE COMPLETED

State the principal requirements, explain the trade-offs between them and identify which claims still need stronger evidence. Add practical implications only where the reviewed literature supports them.

4.1 Limitations and future work

Discuss the coverage of the search, the comparability of evaluation methods and the limits of generalisation. Identify the most useful next research step.

References

- [1] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27, 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html.
- [2] Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html>.

Glossary

CTGAN Conditional tabular generative adversarial network: a GAN-based approach to generating synthetic tabular data using conditional generation.

fidelity The extent to which synthetic data reproduce relevant properties of the target data.

GAN Generative adversarial network: a generative modelling approach in which a generator and a discriminator are trained with competing objectives.

ML Machine learning: methods that learn patterns from data to perform tasks such as prediction or classification.

synthetic data Artificially generated data designed to represent selected properties of observed or target data.

utility The usefulness of data for a specified downstream task and evaluation setting.

A Mathematical notation examples

Template examples only. Adapt the notation and objectives to the research methods actually used.

A.1 Inline and numbered expressions

Let $\mathcal{D}_{\text{syn}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote a synthetic training dataset, where $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector and y_i its target. An illustrative empirical training objective is

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(\mathbf{x}_i), y_i). \quad (1)$$

Here f_{θ} is the model and ℓ is its loss function. Equation (1) demonstrates a clickable equation reference.

A.2 Aligned expressions

For regression, a squared-error loss and an illustrative real-data evaluation risk can be written as

$$\ell(\hat{y}, y) = (\hat{y} - y)^2, \quad (2)$$

$$R_{\text{real}}(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim P_{\text{real}}} [\ell(f_{\theta}(\mathbf{x}), y)]. \quad (3)$$

Training on synthetic data does not, by itself, establish performance on real data; Equation (3) defines a separate evaluation quantity.

A.3 Matrices and cases

A feature matrix and a binary decision rule illustrate two other useful forms:

$$\mathbf{X} = \begin{bmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nd} \end{bmatrix}, \quad \hat{y} = \begin{cases} 1, & p_{\theta}(y = 1 \mid \mathbf{x}) \geq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$